



University of Sadat City
Faculty of Computers and Artificial Intelligence (FCAI)
Course: Machine Learning
Lab Exam



Team Name	Grade_Predictor
Project Title	Student_Performance_Grade_Prediction

No.	Student Name
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Dataset description

The dataset contains **10,000 student records** with 12 features including demographic info (gender, race/ethnicity, parental education level), academic conditions (lunch type, test preparation course), and subject scores (math, reading, writing, science). The target variable is grade (A, B, C, D, F) derived from the total score.

Goal of Project

The goal is to **predict a student's grade** (A/B/C/D/F) based on their demographic background and subject scores, using Machine Learning classification algorithms to identify the key factors that affect academic performance.

Data preprocessing

- Removed **roll_no** column (irrelevant identifier)
- Converted score columns to numeric, replacing invalid values with NaN
- **Dropped rows with missing values** (dropna)
- Removed **outliers** from score columns using IQR method (lower = $Q1 - 3 \times IQR$, upper = $Q3 + 3 \times IQR$)
- Encoded categorical columns (gender, race_ethnicity, parental_level_of_education, grade) using **LabelEncoder**



Feature Selection

Features used:

gender, race_ethnicity, parental_level_of_education, lunch, test_preparation_course, math_score, reading_score, writing_score, science_score (Dropped total_score and grade from features to avoid data leakage)

Data Scaling

Applied **StandardScaler** on training data (fit_transform) and test data (transform) to normalize feature values.

Data Split

Split: **80% training / 20% testing** using train_test_split with random_state=23 and stratify=y to maintain class balance.

ML Algorithms

- **Decision Tree Classifier**
- **Random Forest Classifier**
- **K-Nearest Neighbors (KNN)**

Evaluation metrics

- **Accuracy** – overall correct predictions
- **Precision** – weighted average precision per class
- **Recall** – weighted average recall per class
- **Confusion Matrix** – per-class prediction breakdown
- **Classification Report** – full per-class metrics

Conclusion (Result Analysis)

The three models were compared based on Accuracy, Precision, and Recall. **Random Forest** achieved the best performance among the three, as ensemble methods generally outperform single-tree and distance-based models on tabular data with mixed feature types. Decision Tree was simpler but prone to overfitting, while KNN was sensitive to feature scaling.



Future Direction (How to enhance our model in the future)

- Apply **hyperparameter tuning** (GridSearchCV) for better model optimization
- Try advanced models like **Gradient Boosting (XGBoost)** or **SVM**
- Use **feature importance** from Random Forest to reduce dimensionality
- Handle class imbalance using **SMOTE** if needed
- Deploy the model as a **web application** for real-time grade prediction

Drive URL (.ppt + Python Code)

[Drive Link](#)

References

1. Dataset:

[Kaggle – Student Performance Dataset](#)

2. Scikit-learn: Machine Learning in Python

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4. Visualization:

Hunter, J. D. (2007). *Matplotlib: A 2D Graphics Environment*. Computing in Science & Engineering, 9(3), 90–95.

5. Random Forest Paper:

Breiman, L. (2001). *Random Forests*. Machine Learning, 45, 5–32.

6. KNN Reference:

Cover, T., & Hart, P. (1967). *Nearest neighbor pattern classification*. IEEE Transactions on Information Theory, 13(1), 21–27.

7. Decision Tree Paper:

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